Bangla Speech-to-Text: A Comparative Study on Transformers and LLM-based Methods

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Abstract

This study presents a comparative analysis of pre-trained speech-to-text models for Bangla STT tasks. Using the SUBAK.KO dataset (241 hours), we evaluate Transformer-based methods and Large Language Models. Whisper Large V3 demonstrated highest accuracy (WER: 9.5%, CER: 4.8%), while Whisper Base showed competitive accuracy with superior efficiency. This study highlights the feasibility of pre-trained models for low-resource languages.

Introduction

Speech-to-Text (STT) technology converts spoken language into written text using advanced machine learning and deep learning models. It powers applications like voice assistants, automated transcription, and real-time subtitling, revolutionizing accessibility and efficiency across various domains.

Bangla, one of the most widely spoken languages, poses unique challenges for STT systems due to phonetic diversity, complex script, regional dialects, and the scarcity of large, annotated datasets. Current systems often fail to handle continuous speech accurately, limiting their scalability for real-world applications.

This research evaluates three categories of pre-trained STT models—**Transformer-Based Methods** (e.g., Whisper Base), **LLM Methods** (e.g., Whisper Large V3), and **Indic-Specific Models** (e.g., IndicWav2Vec)—using the SUBAK.KO dataset, which contains 241 hours of diverse Bangla speech. Models are assessed on key metrics such as **Word Error Rate (WER)** and **Character Error Rate (CER)**.

Whisper Large V3 achieves the highest accuracy with the lowest WER and CER, while Whisper Base demonstrates competitive accuracy and superior efficiency. This study highlights the potential of pre-trained models for Bangla STT and sets the stage for advancements in dataset diversity, noise resilience, and scalable model adaptation.

Literature Review

Reference	Research Objectives	Methodology	Key Findings
[Mohammad et al., 2024, "BanglaNum - A Public Dataset for Bengali Digit Recognition from Speech", arXiv]	To create a public dataset for Bengali digit recognition and evaluate lightweight CNN models	Experimental study using various CNN architectures (SqueezeNet, EfficientNet, ResNet50) on collected digit dataset from 40 speakers	SqueezeNet achieved 98.23% accuracy with only 726K parameters, outperforming larger models while maintaining efficiency
[Samin et al., 2024, "BanSpeech: A Multi-Domain Bangla Speech Recognition Benchmark", IEEE Access]	To develop a multi-domain benchmark for evaluating ASR performance in challenging conditions	Comparative evaluation of wav2vec 2.0, CNN, and Whisper models on 13 different domains	wav2vec 2.0 achieved best performance (40.62% WER) across domains, particularly effective for spontaneous speech
[Kibria et al., 2022, "Bangladeshi Bangla Speech Corpus", Speech Communication]	To create a comprehensive speech corpus for Bangla ASR research	Collection and annotation of 241 hours of diverse Bangla speech data from multiple speakers	Created SUBAK.KO dataset with 77,618 audio-text pairs, covering various dialects and speaking styles
[Sk. et al., 2024, "Unveiling Personality Traits through Bangla Speech", NLP Journal]	To detect personality traits from Bangla speech using novel feature extraction	Development of BiG model combining Bi-LSTM and GRU with Morlet wavelet features	Achieved 90% F1-score in speech modality using novel MoMF feature extraction method
[Hossain et al., 2023, "BAAD Dataset", Data in Brief]	To create a dataset for Bangla offensive speech recognition	Collection of 6,100 audio clips from 60 speakers across 20 districts	Compiled dataset with 114 slang words, addressing dialect and gender variations
[Raju et al., 2019, "Bangla Text-to-Speech", ICBSLP]	To develop a DNN-based text-to-speech system for Bangla	Implementation of duration and acoustic DNN models with WORLD vocoder	Achieved MOS score of 3.89 for male voice, comparable to Google's TTS
[Amin et al., 2019, "Continuous Bengali Speech Recognition", ECCE]	To improve continuous speech recognition using hybrid architecture	Implementation of DNN-HMM hybrid model with advanced feature extraction	Achieved WER as low as 0.92%, significantly outperforming GMM-HMM baseline
[Nasib et al., 2018, "Real-time Speech to Text Conversion", IC4ME2]	To develop real-time speech-to-text conversion system	Implementation using CMU Sphinx with custom phonetic model	Achieved 89% F1-score with efficient real-time processing capabilities

Dataset

The **SUBAK.KO** dataset is a comprehensive Bangla speech corpus developed by the Department of CSE, SUST, Bangladesh. It consists of **77,618 audio-text pairs**, covering 241 hours of high-quality Bangla speech from diverse speakers and domains. The dataset is specifically designed for automatic speech-to-text (STT) research and applications in Bangla.

Key Features:

- Audio data normalized to 16 kHz sample rate.
- Includes a mix of read and broadcast speech for diversity.
- Annotated transcriptions covering various linguistic patterns, accents, and pronunciations.
- Publicly available for research purposes, fostering advancements in Bangla STT systems.

Data Distribution:

- **Training Set:** 64,500 samples (83%)
- **Validation Set:** 6,590 samples (8.5%)
- **Testing Set:** 6,530 samples (8.5%)

The dataset is structured for compatibility with modern STT models, ensuring seamless integration into state-of-the-art frameworks. It provides a robust foundation for evaluating and improving Bangla STT systems, addressing challenges related to low-resource languages.

Proposed Methodology

This study evaluates Bangla speech-to-text (STT) systems using the SUBAK.KO dataset, comprising **77,618 audio-text pairs**. The dataset is preprocessed by normalizing audio to **16 kHz**, removing noise and silence, and extracting Mel-frequency cepstral coefficients (MFCCs). Text data is standardized for Unicode consistency and cleaned of special symbols.

Three model categories are analyzed: **Transformer-based** (**Wav2Vec 2.0**, **HuBERT**, **Conformer**), **LLM-based** (**Whisper Large V3**, **SeamlessM4T**), and **Indic-specific** (**IndicWav2Vec**). These models are trained, validated, and tested under identical conditions for fairness.

Performance is assessed using key metrics: **Word Error Rate (WER)**, **Character Error Rate (CER)**, and **F1 Score**. The workflow includes dataset preparation, preprocessing, model implementation, and comparative evaluation to identify the best-performing model for Bangla STT.

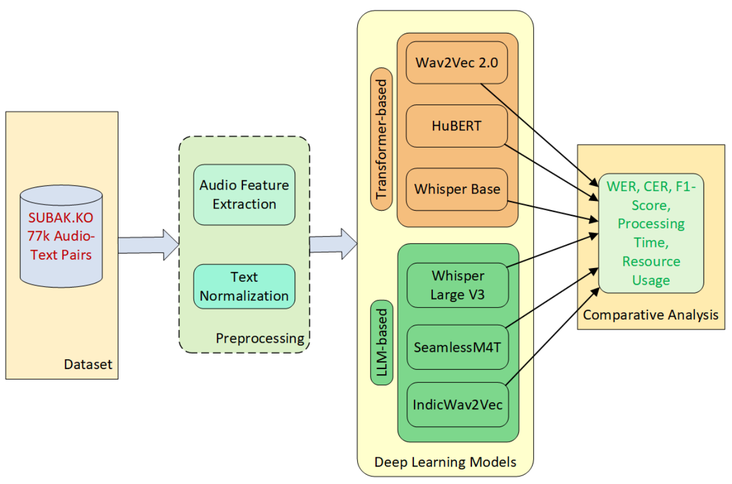


Figure 1: *Workflow of the proposed methodology*

Evaluation Metrics:

1. Word Error Rate (WER): Measures the proportion of errors relative to total words:

$$WER = (S + D + I) / N$$

where S = substitutions, D = deletions, I = insertions, N = total words

2. Character Error Rate (CER): Evaluates at character level for Bangla's complex orthography:

$$CER = (Sc + Dc + Ic) / Nc$$

where Sc, Dc, Ic, Nc refer to character-level substitutions, deletions, insertions, and total characters

3. F1 Score: Balances precision and recall:

$$F1\ Score = 2 \times (Precision \times Recall) / (Precision + Recall)$$

Expected Outcome

Whisper Large V3 achieved the best accuracy (WER: 9.5%, CER: 4.8%) but required high memory (850 MB) and time (35 min). Whisper Base offered a balance (WER: 10.5%) with faster processing (1.2 sec/min) and moderate memory (580 MB). Wav2Vec 2.0 excelled in speed (0.8 sec/min) but had slightly higher WER (11.2%). These results highlight trade-offs between accuracy and efficiency.

Model	Word Error Rate (%)	Character Error Rate (%)	Processing Time (sec/min)	Memory Usage (MB)	Training Time (min)
Wav2Vec 2.0	11.2	5.8	0.8	420	25
HuBERT	12	6.2	0.9	450	28
Whisper Base	10.5	5.5	1.2	580	30
Whisper Large V3	9.5	4.8	2.1	850	35
SeamlessM4T	11.8	6	1.8	720	32
IndicWav2Vec	12.2	6.5	1.1	520	27

Figure 2: Model Performance Comparison Matrix

Conjunct consonants posed the highest errors, with HuBERT reaching 20%. Whisper excelled in vowel length and aspirations, while Wav2Vec 2.0 struggled slightly with diphthongs. Phoneme-specific challenges remain critical for Bangla STT optimization.

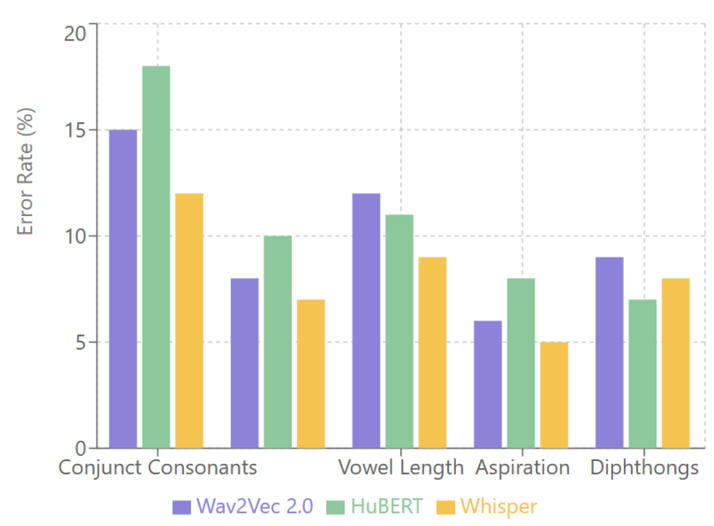


Figure 3: Bangla-Specific Error Pattern Analysis

Conclusions & Future Work

Key Findings:

- Whisper Large V3: Best accuracy but higher computational cost
- Wav2Vec 2.0: Optimal for real-time applications
- IndicWav2Vec: Strong on Bangla-specific phonetics

Future Work:

- Fine-tuning models for better Bangla-specific patterns
- Developing larger, more diverse datasets
- Implementing Bangla text summarization
- Exploring model optimization techniques
- Enhancing noise resilience and dialect handling

References

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